

Data Cleaning and Preparation

Introduction

Data cleaning and preparation are essential steps in the data analysis process. Real-world datasets often contain missing values, duplicate records, and inconsistent formats that can affect the accuracy of analysis. This project focuses on cleaning and preparing a marketing campaign dataset by identifying missing values, handling inconsistencies, checking for duplicates, and converting data into appropriate formats. The objective is to improve data quality and make the dataset ready for further analysis.

Dataset Overview

The dataset used in this project is a marketing campaign dataset containing customer information such as age, education, marital status, income, purchase behavior, and customer enrollment date. The dataset includes multiple attributes that can be used for customer analysis and business decision-making. Before analysis, the dataset was examined for missing values, duplicate records, and formatting issues.

Data Cleaning Process

The dataset was checked for missing values using Python. It was observed that the Income column contained 24 missing values. These

missing values were replaced with the mean value of the Income column to maintain data consistency.

...		0
ID		0
Year_Birth		0
Education		0
Marital_Status		0
Income		24
Kidhome		0
Teenhome		0
Dt_Customer		0
Recency		0
MntWines		0
MntFruits		0
MntMeatProducts		0
MntFishProducts		0
MntSweetProducts		0
MntGoldProds		0
NumDealsPurchases		0

Duplicate Records Check

The dataset was examined for duplicate records using the duplicated() function. No duplicate records were found in the dataset.

[14]  df.duplicated().sum()

✓ Os

▼ ... np.int64(0)

[15] df.info()

✓ Os

▼ <class 'pandas.core.frame.DataFrame'>
RangeIndex: 2240 entries, 0 to 2239
Data columns (total 29 columns):
Column Non-Null Count Dtype
--- -
0 ID 2240 non-null int64
1 Year_Birth 2240 non-null int64
2 Education 2240 non-null object
3 Marital_Status 2240 non-null object
4 Income 2240 non-null float64
5 Kidhome 2240 non-null int64
6 Teenhome 2240 non-null int64
7 Dt_Customer 2240 non-null object
8 Recency 2240 non-null int64
9 MntWines 2240 non-null int64
10 MntFruits 2240 non-null int64
11 MntMeatProducts 2240 non-null int64
12 MntFishProducts 2240 non-null int64
13 MntSweetProducts 2240 non-null int64
14 MntGoldProds 2240 non-null int64
15 NumDealsPurchases 2240 non-null int64
16 NumWebPurchases 2240 non-null int64
17 NumCatalogPurchases 2240 non-null int64
18 NumStorePurchases 2240 non-null int64
19 NumWebVisitsMonth 2240 non-null int64

19 NumWebVisitsMonth 2240 non-null int64
20 AcceptedCmp3 2240 non-null int64
21 AcceptedCmp4 2240 non-null int64
22 AcceptedCmp5 2240 non-null int64
23 AcceptedCmp1 2240 non-null int64
24 AcceptedCmp2 2240 non-null int64
25 Complain 2240 non-null int64
26 Z_CostContact 2240 non-null int64
27 Z_Revenue 2240 non-null int64
28 Response 2240 non-null int64

dtypes: float64(1), int64(25), object(3)

memory usage: 507.6+ KB

Data Format Correction

The Dt_Customer column was converted from text format to datetime format using pandas. This ensures that date-related analysis can be performed accurately.

Dt_Customer	
0	2012-09-04
1	2014-03-08
2	2013-08-21
3	2014-02-10
4	2014-01-19

dtype: datetime64[ns]

...	Dt_Customer	
	0	2012-09-04
	1	2014-03-08
	2	2013-08-21
	3	2014-02-10
	4	2014-01-19
	5	2013-09-09
	6	2012-11-13
	7	2013-05-08
	8	2013-06-06
	9	2014-03-13

dtype: datetime64[ns]

Results

- Missing values in the Income column were successfully handled.
- No duplicate records were found in the dataset.
- The customer date column was converted to datetime format.
- The dataset was cleaned and prepared for future analysis.

Conclusion

The data cleaning and preparation process improved the quality and reliability of the marketing campaign dataset. Missing values were handled, duplicate records were verified, and date formatting issues were corrected. The cleaned dataset is now suitable for further analysis and machine learning applications.